Differential recognition of oligomannose isomers by glycan-binding proteins involved in innate...

Search within citing articles

The mannose receptor ligands and the macrophage glycome

RD Cummings - Current opinion in structural biology, 2022 - Elsevier

A unique glycan-binding protein expressed in macrophages and some types of other immune cells is the mannose receptor (MR, CD206). It is an endocytic, transmembrane …

Save Cite Cited by 146 Related articles All 5 versions

Insights into glycobiology and the protein-glycan interactome using glycan microarray technologies

J Heimburg-Molinaro, AY Mehta, CA Tilton… - Molecular & Cellular …, 2024 - mcponline.org

Glycans linked to proteins and lipids and also occurring in free forms have many functions, and these are partly elicited through specific interactions with glycan-binding proteins …

Save Cite Cited by 32 Related articles All 6 versions

Advances in Fragmentation Techniques for Glycomics and Glycoproteomics

V Sandilya, S Sahioun, TA Suma… - Mass Spectrometry …, 2026 - Wiley Online Library

Glycosylation, the enzymatic addition of carbohydrate moieties to proteins, is essential for immune recognition, protein folding, and disease progression. The structural complexity of …

Save Cite Cited by 7 Related articles All 4 versions

Serum *N*-Glycan Profiling Identifies Candidate Glycan Biomarkers for Early Detection and Prediction of Alzheimer's Disease

S Onigbinde, J Solomon… - Journal of proteome …, 2025 - ACS Publications

Alzheimer's disease (AD) is a neurodegenerative disorder marked by progressive cognitive decline, affecting millions worldwide. Early diagnosis and intervention are crucial yet …

Save Cite Cited by 11 Related articles All 8 versions

LectinOracle: a generalizable deep learning model for lectin–glycan binding prediction

J Lundstrøm, E Korhonen, F Lisacek… - Advanced …, 2022 - Wiley Online Library

Ranging from bacterial cell adhesion over viral cell entry to human innate immunity, glycan-binding proteins or lectins are abound in nature. Widely used as staining and …

Save Cite Cited by 70 Related articles All 15 versions

O-glycosylation in viruses: A sweet tango

A Ming, J Zhao, Y Liu, Y Wang, X Wang, J Li, L Zhang - Mlife, 2024 - Wiley Online Library

O-glycosylation is an ancient yet underappreciated protein posttranslational modification, on which many bacteria and viruses heavily rely to perform critical biological functions involved …

Save Cite Cited by 19 Related articles All 13 versions

Analysis of carbohydrates and glycoconjugates by matrix-assisted laser desorption/ionization mass spectrometry: An update for 2021–2022

DJ Harvey - Mass Spectrometry Reviews, 2024 - pmc.ncbi.nlm.nih.gov

The use of matrix-assisted laser desorption/ionization (MALDI) mass spectrometry for the analysis of carbohydrates and glycoconjugates is a well-established technique and this …

Save Cite Cited by 16 Related articles All 9 versions

N-glycosylation of mannose receptor (CD206) regulates glycan binding by C-type lectin domains

K Stavenhagen, AY Mehta, L Laan, C Gao… - Journal of Biological …, 2022 - jbc.org

The macrophage mannose receptor (MR, CD206) is a transmembrane endocytic lectin receptor, expressed in selected immune and endothelial cells, and is involved in immunity …

Save Cite Cited by 34 Related articles All 5 versions

Absolute affinities from quantitative shotgun glycomics using concentration-independent (COIN) native mass spectrometry

DT Bui, J Favell, EN Kitova, Z Li… - ACS Central …, 2023 - pmc.ncbi.nlm.nih.gov

Native mass spectrometry (nMS) screening of natural glycan libraries against glycan-binding proteins (GBPs) is a powerful tool for ligand discovery. However, as the glycan …

☆ Save 99 Cite Cited by 27 Related articles All 6 versions

Trends in oligomannosylation and α1, 2-mannosidase expression in human cancers

S Chatterjee, J Ugonotti, LY Lee, A Everest-Dass… - …, 2021 - pmc.ncbi.nlm.nih.gov

Aberrant protein glycosylation is a prominent cancer feature. While many tumour-associated glycoepitopes have been reported, advances in glycoanalytics continue to uncover new …

☆ Save 99 Cite Cited by 46 Related articles All 11 versions